



CODE: PCC-CS-601
SUBJECT NAME: INTELLIGENT SYSTEMS
NO OF CREDITS: 3

B.TECH 6 th SEMESTER	SESSIONAL:	25
L T P	THEORY EXAM:	75
3 0 0	TOTAL :	100

Pre-requisites: Basics of Data Structures and Mathematics

Course Objectives:

The aim of the course is to introduce to the field of Artificial Intelligence (AI) with emphasis on its use to solve real world problems for which solutions are difficult to express using the traditional algorithmic approach. It explores the essential theory behind methodologies for developing systems that demonstrate intelligent behaviour including dealing with uncertainty, learning from experience and following problem solving strategies found in nature.

MODULE-1:

Biological foundations to intelligent systems I: Artificial neural networks, Backpropagation networks, Radial basis function networks, and recurrent networks.

MODULE-2:

Biological foundations to intelligent systems II: Fuzzy logic, knowledge Representation and inference mechanism, genetic algorithm, and fuzzy neural networks.

MODULE-3:

Search Methods Basic concepts of graph and tree search. Three simple search methods: breadth-first search, depth-first search, iterative deepening search. Heuristic search methods: best-first search, admissible evaluation functions, hill climbing search. Optimisation and search such as stochastic annealing and genetic algorithm.

MODULE-4:

Knowledge representation and logical inference Issues in knowledge representation. Structured representation, such as frames, and scripts, semantic networks and conceptual graphs. Formal logic and logical inference, Knowledge-based systems structures, its basic components. Ideas of Blackboard architectures.

MODULE-5:

Reasoning under uncertainty and Learning Techniques on uncertainty reasoning such as Bayesian reasoning, Certainty factors and Dempster-Shafer Theory of Evidential reasoning, A study of different learning and evolutionary algorithms, such as statistical learning and induction learning.

Course Outcomes:

1. Able to Demonstrate knowledge of the fundamental principles of intelligent systems
2. able to analyse and compare the relative merits of a variety of AI problem solving techniques



REFERENCES:

1. Luger G.F. and Stubblefield W.A. (2008). **Artificial Intelligence: Structures and strategies for Complex Problem Solving**. Addison Wesley, 6th edition.
2. Russell S. and Norvig P. (2009). **Artificial Intelligence: A Modern Approach**. Prentice-Hall, 3rd edition.